DR. SHYAMA PRASAD MUKHERJEE UNIVERSITY, RANCHI

**MASTER OF COMPUTER APPLICATION**

MID SEMESTER – III EXAMINATION – 2026

TIME: 1 HOUR — FULL MARKS:20

Sub: Analysis & Design of Algorithm — Paper: CCMCA-304

**Group-A**

1. Choose the correct answer from the following questions: 5 x 1 = 20

A. In greedy method which type of solution is generated________________

a. Optimal solution b. Worst solution

c. Best solution d. All solutions

B. The recursive versions of binary search use a ______________ structure.

a. Branch and bound b. Dynamic programming

c. Divide and conquer d. Simple recursive

C. Which of the following algorithms is used to find the shortest path between two vertices in a graph?

a. Breadth First Search b. Depth First Search

c. Dijkstra's shortest path algorithm d. Bellman-Ford algorithm

D. To main measures of the efficiency of an algorithm are

a. Time & Space complexity b. Data & Space

c. Processor & Memory d. Complexity & capacity

E. Fractional knapsack problem is also known as

a. Continuous knapsack problem b. Divisible knapsack problem

c. 0/1 knapsack problem d. Non continuous knapsack problem

**Group-B**

Answer any THREE of the following: 3 x 5 = 15

2. How can multiply large integers using Divide and Conquer Process by Karatsuba Algorithm? Explain with suitable example.
3. Explain the chained matrix multiplication with suitable example.
4. Explain General method of Greedy method. Find the greedy solution for following job sequencing with deadlines problem n = 7, (p1, p2, p3, p4, p5, p6, p7) = (3,5,20,18,1,6,30), (d1, d2, d3, d4, d5, d6, d7) = (1,3,4,3,2,1,2).
5. What is back tracking? Where Back tracking is used to solve the problem. Explain with suitable example.
6. Find the optimal solution of the Knapsack instance n = 7, M = 15, (p1, p2, p3, p4, p5, p6, p7) = (10,5,15,7,6,18,3) and (w1, w2, w3, w4, w5, w6, w7) = (2,3,5,7,1,4,1).

xxxxxxxxxxxxxxxx

Scanned with OKEN Scanner